

MTH001- Elementary Mathematics

Solved Final Term Papers For Final Term Exam Preparation

Question No: 1 (Marks: 1) - Please choose one

The difference between the upper and the lower class boundaries of a class are known as:

- ▶ Class Marks
- ▶ **Class Interval**
- ▶ Class frequency

Question No: 2 (Marks: 1) - Please choose one

The probability of sure event is

- ▶ 0
- ▶ **1**
- ▶ 0.5

Question No: 3 (Marks: 1) - Please choose one

What is the simple interest earned on Rs.3000 invested at 8% per annum for 6 months?

- ▶ **$3000 \times 6 \times 0.08$**
- ▶ $3000 \times 60 \times 0.08$
- ▶ $3000 \times 0.5 \times 0.08$
- ▶ $3000 \times 1.5 \times 0.08$

Question No: 4 (Marks: 1) - Please choose one

The probability of vowel letters from the word STATISTICS is

▶ 2/10

▶ 3/10

▶ 0

▶ 10/10

Question No: 5 (Marks: 1) - Please choose one

Smoking habits of the residents of an area are

▶ Quantitative

▶ Qualitative

Question No: 6 (Marks: 1) - Please choose one

In a symmetric distribution

▶ The mean, median, and mode are equal

▶ The mean is the largest measure of location

▶ The median is the largest measure of location

▶ The standard deviation is the largest value

Question No: 7 (Marks: 1) - Please choose one

When two coins are tossed simultaneously, the probability of at least one head is _____.

▶ 1/4

▶ 3/4

▶ 1/2

► 1

Question No: 8 (Marks: 1) - Please choose one

20 m long rope is cut to the length of 15m. what is percenrage decrease?

► $(5/20 \times 100)\%$

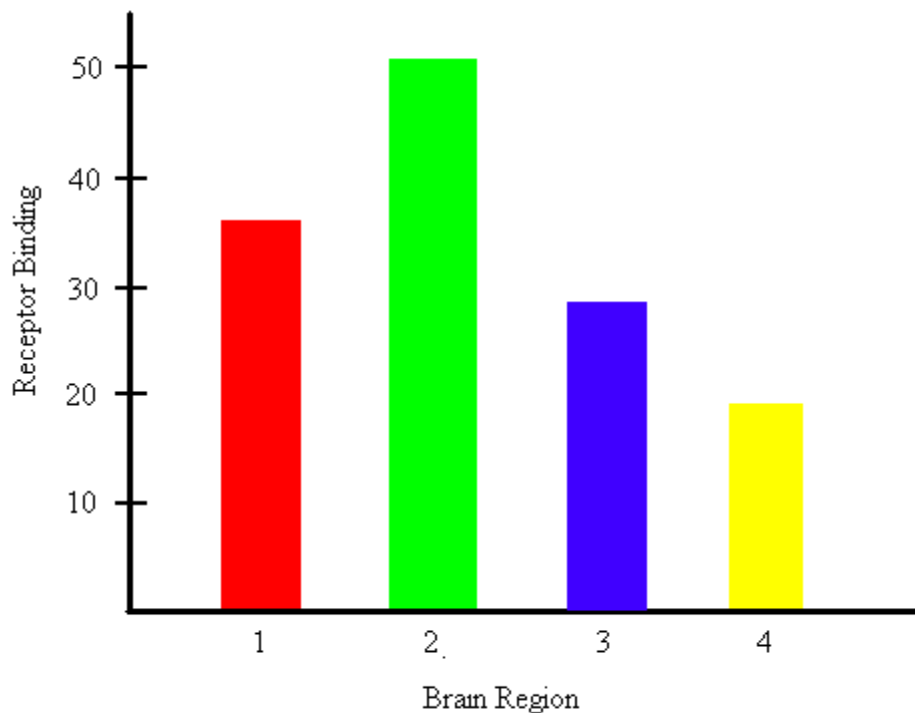
► $(15/20 \times 100)\%$

► $(5/20)\%$

► $(2/2000)\%$

Question No: 9 (Marks: 1) - Please choose one

The chart below is agraph.



► Pie

► **Bar**

► Line

► Quadratic

Question No: 10 (Marks:)

List price of a shirt is Rs 450. If the discount rate is 20%, calculate the discount amount of the shirt.

Question No: 11 (Marks: 1) - Please choose one

Third Quartile = Q3 =

► P3

► Median

► **None of the above**

► Mode

Question No:12 (Marks: 1) - Please choose one

Standard deviation is always measure from

► **Mean**

► Median

► Mode

Question No: 13 (Marks: 1) - Please choose one

The probability of sure event is

► 0

▶ 1

▶ 0.5

Question No: 14 (Marks: 1) - Please choose one

The probability of tail when a coin is tossed is

▶ 0

▶ 1

▶ 1/2

Question No: 15 (Marks: 1) - Please choose one

A major disadvantage of mean is that it is affected by..

▶ Extremely large values

▶ Extremely small values

▶ Middle values

Question No: 16 (Marks: 1) - Please choose one

Mode is the measure which always exists in any numerical data

▶ True

▶ False

Question No: 17 (Marks: 1) - Please choose one

When 3 coins are tossed simultaneously (3 heads) will be.

▶ 3/8

▶ 1/8

▶ 1/4

▶ 1

Question No: 18 (Marks: 1) - Please choose one

If A is the sure event, then $P(A)=$

▶ 1

▶ 1/2

▶ 1/4

▶ 3/4

Question No: 19 (Marks: 1) - Please choose one

Probability of a sure event is

▶ 1/2

▶ 0

▶ 1

▶ 1/4

Question No: 20 (Marks: 1) - Please choose one

If E and F are mutually exclusive events such that $P(E)=0.4$ and $P(F)=0.5$ then.

▶ 0.2

▶ 0.9

▶ 0.1

▶ 0.5

Question No: 21 (Marks: 1) - Please choose one

Which of them is a proposition?

▶ x is greater then 2

▶ $3+7=10$

▶ close the door

▶ what time is it?

Question No:22 (Marks: 1) - Please choose one

Conjunction of two statements p and q is denoted by

▶ $p \wedge q$

▶ $p \vee q$

▶ $p \leftrightarrow q$

▶ $p \rightarrow q$

Question No: 23 (Marks: 1) - Please choose one

The component bar chart should be used when we have available to use in formation regarding totals and their components.

▶ True

▶ False

Question No: 24 (Marks: 1) - Please choose one

A conditional statement is logically equivalent.

▶ contra positive

▶ Inverse

▶ converse to its

Question No: 25 (Marks: 1) - Please choose one

Which one of the following is the negation of the statement

▶ If Tanveer live in Lahore then he does not live in Pakistan

▶ Tanveer does not live in Lahore and he lives in Pakistan

▶ If Tanveer does not live in Lahore then he does not live in Pakistan

▶ Tanveer does not live in Lahore then he lives in Pakistan

Question No: 26 (Marks: 1) - Please choose one

According to Demorgan's Law $\sim (p \wedge q) =$

► $\sim p \wedge q$

► $\sim p \wedge \sim q$

► $\sim p \vee \sim q$

► $p \wedge \sim q$

Question No: 27 (Marks: 1) - Please choose one

Order of elements in a set does not matter

► True

► False

Question No: 28 (Marks: 1) - Please choose one

If $A = \{1, 2, 3, 4\}$ and $B = \{5, 6, 7\}$, then $A \cap B$.

► $\{1, 2, 3\}$

► $\{5, 6, 7\}$

► $\{1, 2, 3, 4, 5, 6, 7\}$



Question No: 29 (Marks: 1) - Please choose one

Which one of the following sets is finite?

▶ set of real numbers

▶ set of even numbers

▶ $\{x \mid x \in \mathbb{Z} \wedge 0 \leq x \leq 12\}$

▶ $\{x \mid x \in \mathbb{R} \wedge x \geq 12\}$

Question No: 30 (Marks: 1) - Please choose one

If $R = \{(a,a), (b,b), (c,c)\}$ is a relation on the set $A = \{a,b,c\}$

▶ symmetric only

▶ symmetric and Reflexive

▶ Equivalence relation

▶ Reflexive only

Question No: 31 (Marks: 1) - Please choose one

$2x + 6y$ is a

▶ Monomial

▶ Binomial

▶ Trinomial

▶ none of these

Question No: 32 (Marks: 1) - Please choose one

Order of matrix $A = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ is

▶ **1 x 1**

▶ 2 x 2

▶ 2 x 1

▶ 1 x 2

Question No: 33 (Marks: 1) - Please choose one

If matrix A is of order 2 x 3 and B is of order 3 x 4, then order of matrix AB will be

▶ **2 x 4**

▶ 2 x 3

▶ 2 x 2

▶ 3 x 2

Question No: 34 (Marks: 1) - Please choose one

Data which is not arranged in ascending or descending order is called

▶ grouped data

▶ **raw data**

▶ qualitative data

Question No: 35 (Marks: 1) - Please choose one

If $x : y = y : 1$ then $x =$

- ▶ y^2
- ▶ y
- ▶ $\frac{y}{x}$
- ▶ $x^2 y$

Question No: 36 (Marks: 1) - Please choose one

If set A has m elements and set B has n elements then the total numbers of relation from A to B are

- ▶ $m \times n$
- ▶ m
- ▶ $2 \times n$
- ▶ $m + n$
- ▶ mn

Question No: 37 (Marks: 1) - Please choose one

Determine the range of the function g from $X = \{2,4,5\}$ to $Y = \{1,2,4,6\}$ defined as $g = \{(2,6),(4,2),(5,1)\}$

- ▶ Range of $g = \{1,2,6\}$
- ▶ Range of $g = \{2,6\}$
- ▶ Range of $g = \{2,4,5\}$

Question No: 38 (Marks: 1) - Please choose one

Suppose A is a set with m elements and B is a set with n elements. How many functions

are there from A to B?

▶ **m.n**

m

▶ n

m m

▶ n . n

Question No: 39 (Marks: 1) - Please choose one

If the price of shoes increase from Rs.600 to 900, what is the percentage increase?

▶ (600/3)%

▶ (900/6)%

▶ (600/9)%

▶ **(300/6)%**

Question No: 40 (Marks: 1) - Please choose one

The value of all assets that are reasonably expected to be converted in to cash with in one year in the normal course of business are called

▶ **Liabilities**

▶ Current assets

▶ Face value

Question No: 41 (Marks:) - Please choose one

Define Observations.

Question No: 42 (Marks:) - Please choose one

A pair of fair dice is rolled once .Find the probability that the sum of the face is 9.

Question No: 43 (Marks:) - Please choose one

let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x+1$

$x+2$

show that f is one - to -one and onto

Question No: 44 (Marks:) - Please choose one

Finde the value of x in the proportion $(5x+1):3 = (2x+2):7$ $(6x):(4x):7$

Question No: 45 (Marks:1) - Please choose one

The chart below is a graph

► Bar

► Pie

► Line

► Simple bar

Question No: 45 (Marks:) - Please choose on

Consider the following diagrams

FINALTERM EXAMINATION
Spring 2010
MTH001- Elementary Mathematics

Ref No: 1735901

Time: 90 min

Marks: 60

Student Info	
Student ID:	
Center:	
Exam Date:	

For Teacher's Use Only									
Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

Question No: 1 (Marks: 1) - Please choose one

Tabulation means, the process of arranging the data into

- ▶ Different classes
- ▶ Rows
- ▶ Columns
- ▶ Rows and Columns

Question No: 2 (Marks: 1) - Please choose one

The difference between the upper and the lower class boundaries of a class are known as:

- ▶ Class Marks
- ▶ **Class Interval**
- ▶ Class frequency

Question No: 3 (Marks: 1) - Please choose one

For the given data 2, 3, 7, 0,-8 G.M will be

- ▶ Negative
- ▶ Positive
- ▶ zero

Question No: 4 (Marks: 1) - Please choose one

The graph of cumulative frequency is called

- ▶ Histogram
- ▶ **Frequency polygon**
- ▶ Ogive

Question No: 5 (Marks: 1) - Please choose one

The probability of sure event is

- ▶ 0
- ▶ **1**
- ▶ 0.5

Question No: 6 (Marks: 1) - Please choose one

The probability of an event always lies between:

- ▶ -1 & +1
- ▶ **-1 & 0**

► 0 & 1

Question No: 7 (Marks: 1) - Please choose one

The probability of vowel letters from the word STATISTICS is

- 2/10
- 3/10
- 0
- 10/10

Question No: 8 (Marks: 1) - Please choose one

$P(A \text{ or } B) = P(A) + P(B)$, if A & B are

- Mutually exclusive events
- **Independent events**
- Not mutually exclusive events

Question No: 9 (Marks: 1) - Please choose one

$\emptyset \in \{x\}$

- **True**

► False

Question No: 10 (Marks: 1) - Please choose one

Variance is the square root of standard deviation.

► True

► False

Question No: 11 (Marks: 1) - Please choose one

Smoking habits of the residents of an area are

► Quantitative

► Qualitative

Question No: 12 (Marks: 1) - Please choose one

Results obtained by counting are

► Discrete data

► Continuous data

Question No: 13 (Marks: 1) - Please choose one

The geometric mean of 2,8,4 is

- ▶ 2
- ▶ 4
- ▶ 8
- ▶ 0

Question No: 14 (Marks: 1) - Please choose one

Let $A = \{x : x \in N, 4 + x = 3\}$ where N is the set of natural numbers.

Which of the following is tabular form of A ?

- ▶ $\{ \}$
- ▶ $\{1\}$
- ▶ $\{-1\}$
- ▶ $\{3, 4\}$

Question No: 15 (Marks: 1) - Please choose one

a% of b is b% of

- ▶ a
- ▶ 100a

$$\frac{a}{100}$$



$$\frac{b}{100}$$



Question No: 16 (Marks: 1) - Please choose one

The text concatenation operator is used to

- ▶ **combine two text strings.**
- ▶ include “:” and “,”
- ▶ calculate exponentiation: ^
- ▶ make comparisons.

Question No: 17 (Marks: 1) - Please choose one

In a symmetric distribution

- ▶ The mean, median, and mode are equal
- ▶ The mean is the largest measure of location
- ▶ The median is the largest measure of location
- ▶ The standard deviation is the largest value

Question No: 18 (Marks: 1) - Please choose one

When two coins are tossed simultaneously, the probability of at least one head is _____.

- ▶ 1/4
- ▶ 3/4
- ▶ 1/2
- ▶ 1

Question No: 19 (Marks: 1) - Please choose one

For two mutually exclusive events A and B, $P(A) = 0.2$ and $P(B) = 0.4$, then $P(A \cup B)$ = -----.

- ▶ 0.8
- ▶ 0.2
- ▶ 0.6
- ▶ 0.52

Question No: 20 (Marks: 1) - Please choose one

$A = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$ is an example of

- ▶ Row matrix .
- ▶ Column matrix.
- ▶ Square matrix .

- ▶ Identity matrix.

Question No: 21 (Marks: 1) - Please choose one

When a scalar k is multiplied with any matrix then

- ▶ k is multiplied with each row of that matrix.
- ▶ k is multiplied with each column of that matrix.
- ▶ **k is multiplied with all elements of that matrix .**
- ▶ None of these.

Question No: 22 (Marks: 1) - Please choose one

$0.20\% = \dots\dots\dots$

- ▶ 0.020
- ▶ 0.0020
- ▶ 0.00020
- ▶ 0.000020

Question No: 23 (Marks: 1) - Please choose one

20 m long rope is cut to the length of 15m. what is percentage decrease?

- ▶ 25%
- ▶ 33.3%
- ▶ 40%
- ▶ 45%

Question No: 24 (Marks: 1) - Please choose one

If the Amount of Annuity is 10,000 Rs and Accumulation Factor is 26.24, what will be Accumulation Value?

- ▶ 26240
- ▶ **262400**
- ▶ 2624000
- ▶ 2624.0

Question No: 25 (Marks: 1) - Please choose one

Suppose that you are spending 2500 dollars in an investment and Accumulated factor is 20 then Accumulated Value will be

- ▶ 20000

▶ 30000

▶ 40000

▶ 50000

Question No: 26 (Marks: 1) - Please choose one

Monthly installments of a leased car are calculated by using the techniques of

▶ Simple interest method

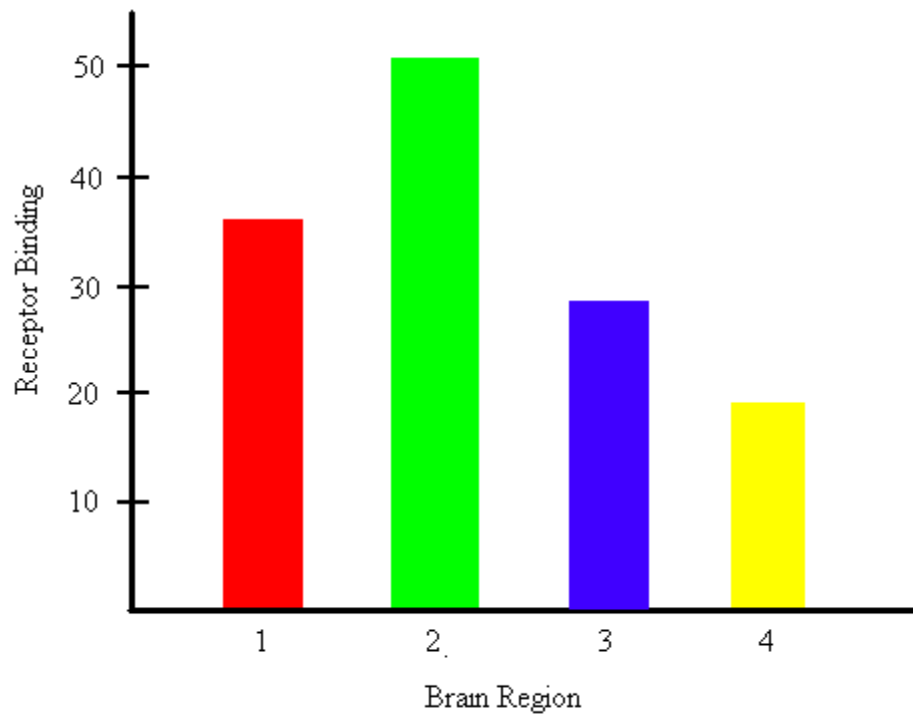
▶ Compound interest method

▶ **Annuity**

▶ Hit & trial method

Question No: 27 (Marks: 1) - Please choose one

The chart below is agraph.



- ▶ Pie
- ▶ **Bar**
- ▶ Line
- ▶ quadratic

Question No: 28 (Marks: 1) - Please choose one

The arrow " $\Leftrightarrow$ " is the conditional operator.

- ▶ True
- ▶ **False**

Question No: 29 (Marks: 1) - Please choose one

Union of any two sets satisfies commutative law.

► True

► False

Question No: 30 (Marks: 1) - Please choose one

The term symmetric and antisymmetric is opposite.

► True

► False

Question No: 31 (Marks: 2)

How many 3-digit numbers can be formed by using each one of the digits 2,3,5,7,9 only once?

Question No: 32 (Marks: 2)

Find Geometric Mean of 2, 4 and 8.

Question No: 33 (Marks: 2)

List price of a shirt is Rs 450. If the discount rate is 20%, calculate the discount amount of the shirt.

Question No: 34 (Marks: 3)

The net price of a computer table after trade discount 35.5% is 2500. Find the list price of the computer table.

Question No: 35 (Marks: 3)

Find D_2 (Second Deciles), D_7 (Seventh Deciles) from the following marks obtained by 20 students on a test in Mathematics.

20 28 29 30 36 37 39 42 53 54 55 58 61 67 68 70 74 81 82 93

Question No: 36 (Marks: 3)

- a) In how many ways can 3 seats on a sofa set be occupied by 5 persons?
- b) In how many ways can six different books be arranged on a shelf?

Question No: 37 (Marks: 5)

Mr. Naveed invested a sum of money five years ago in an account which pays 8% per year compounded semi-annually. The account now contains \$400. What did Mr. Naveed invest five years ago?

Question No: 38 (Marks: 5)

What is the difference between mutually exclusive and exhaustive events? Explain with example.

Question No: 39 (Marks: 5)

Find Harmonic Mean of 3, 5, 6, 6, 7, 10 and 12.